Supplementary Information

Kin group Gini: Extended family and economic inequality in the U.S.

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Appendix A: Comparison with External Estimates

Appendix A1. Comparison of income Gini coefficients

Year	FRED	U.S. Census	Our Results (All HH)
1969	0.391	0.349	0.401
1979	0.404	0.365	0.398
1989	0.431	0.401	0.439
1999	0.458	0.429	0.465
2009	0.468	0.443	0.464
2019	0.484	0.454	0.470

Appendix A2. Comparison of wealth Gini coefficients with external estimates

Year	SCF	PSID	Our Results
1983	0.760		
1984		0.739	0.755
1989	0.781	0.742	0.751
1992	0.770		
1994		0.732	0.749
2003		0.810	0.786
2004	0.809		
2007		0.830	0.801
2009		0.890	0.833
2010	0.846		
2011		0.880	0.811

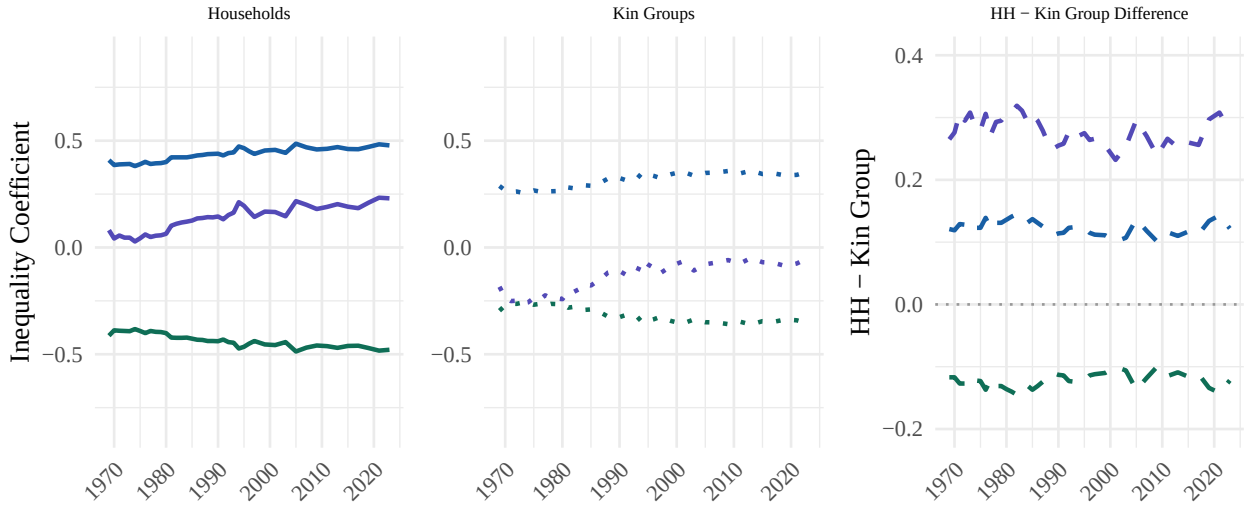
Appendix A compares Gini coefficients at the household level calculated for income and wealth with Gini coefficients from other researchers. Gini coefficients in Appendix A are calculated using the full sample of households in the PSID, rather than just the households who belong to a kin group with at least one other household (which are the results presented in the main paper).

Gini coefficients are not available for all years in other sources, so only years available are presented. Gini benchmarks for income rely on estimates from the Federal Reserve Bank of St. Louis (FRED) and the U.S. Census[1,2]. FRED uses data from the World Bank on U.S. households and the Census uses data from the

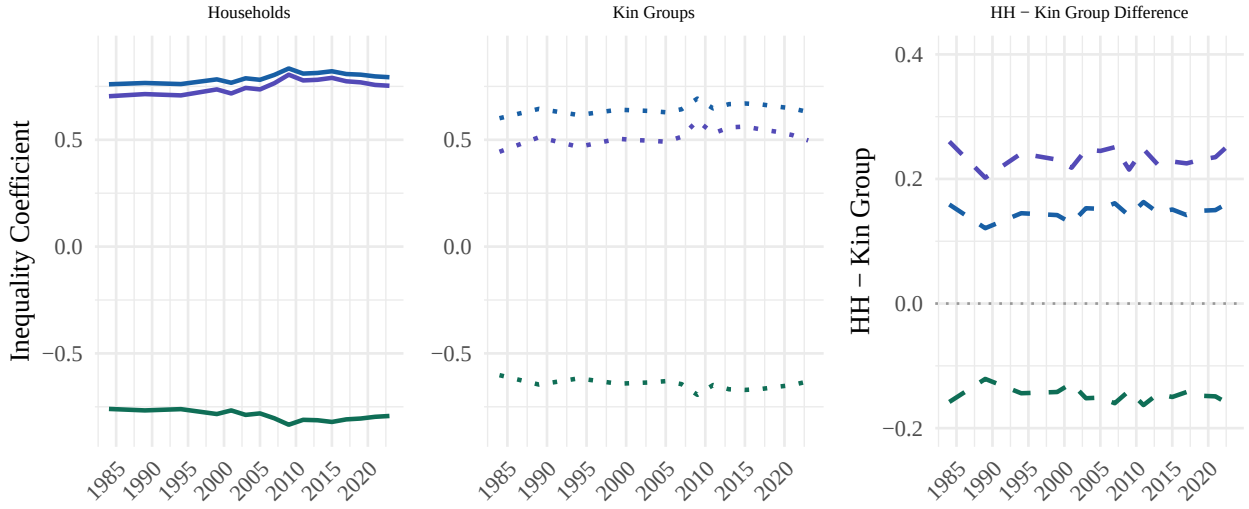
Current Population Survey (CPS). Gini benchmarks for wealth rely on estimates from research papers based on data from the PSID or Survey of Consumer Finances (SCF)[3,4].

Appendix B: Alternative Inequality Measures

Panel A: Income



Panel B: Wealth (excl. home equity)



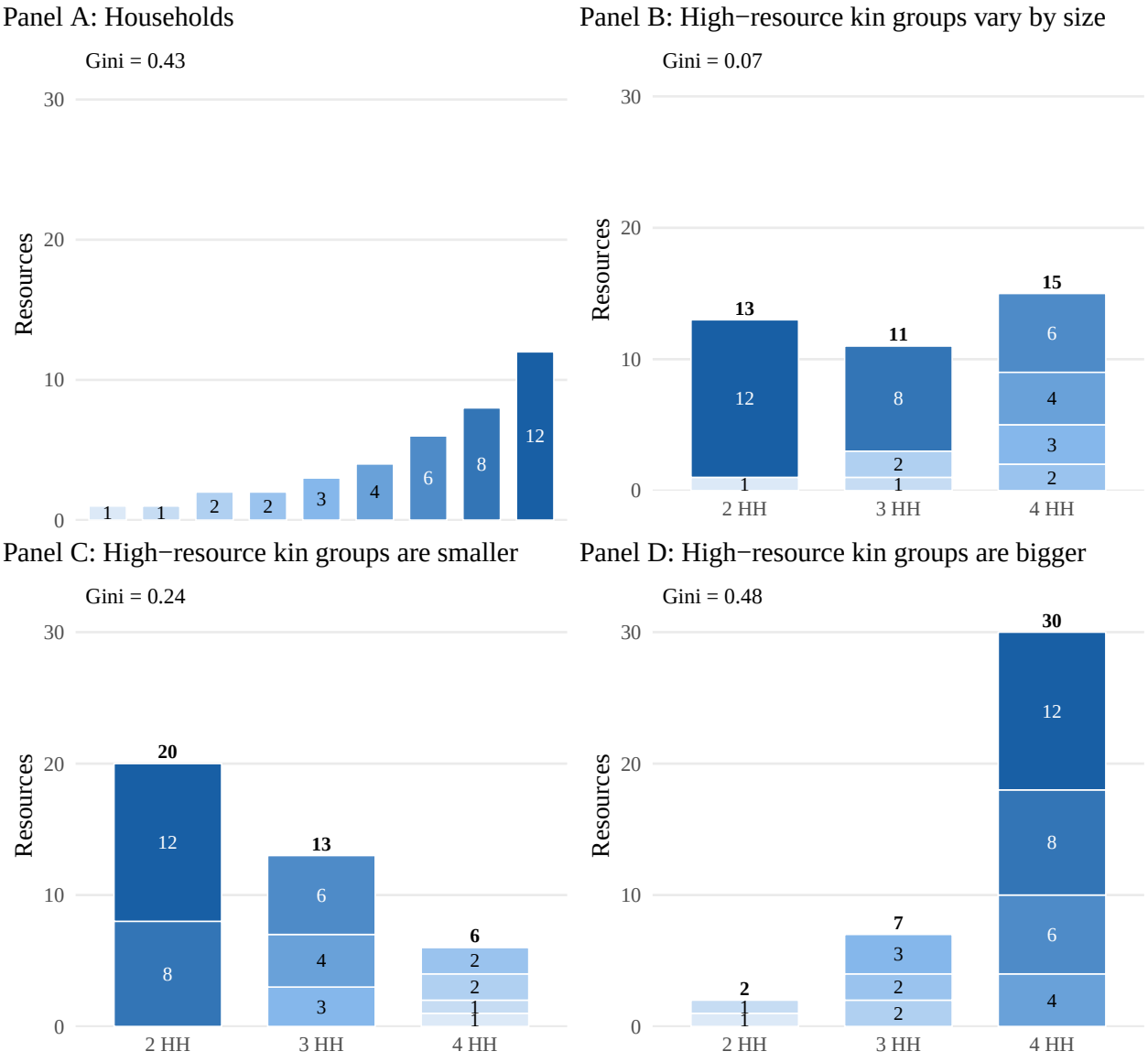
C1 (Bonferroni) C2 (Gini) C3 (Upper Tail)

Appendix B shows three alternative measures of inequality proposed by Aaberge[5]. C1 is the Bonferroni coefficient (greater weight on lower tail); C2 is the standard Gini coefficient; C3 places greater weight on the upper tail of the distribution. All estimates are weighted averages across all available years. Inequality is lower at the kin group level using any measure of inequality.

Appendix C: Simulating Gini Coefficients

Appendix C shows that a kin group-level Gini coefficient is not necessarily lower than a household-level Gini coefficient. The kin group Gini measures inequality between kin groups, comparing the sum of resources held by all households in each kin group. Whether that Gini is higher or lower than the household Gini depends on how high- and low-resource households are distributed across kin groups of different sizes. Appendix C1 illustrates this with a single set of households grouped into kin groups three ways. Appendix C2 uses simulated data to show the most extreme outcomes.

Appendix C1: Illustrating Kin Group Gini Possibilities



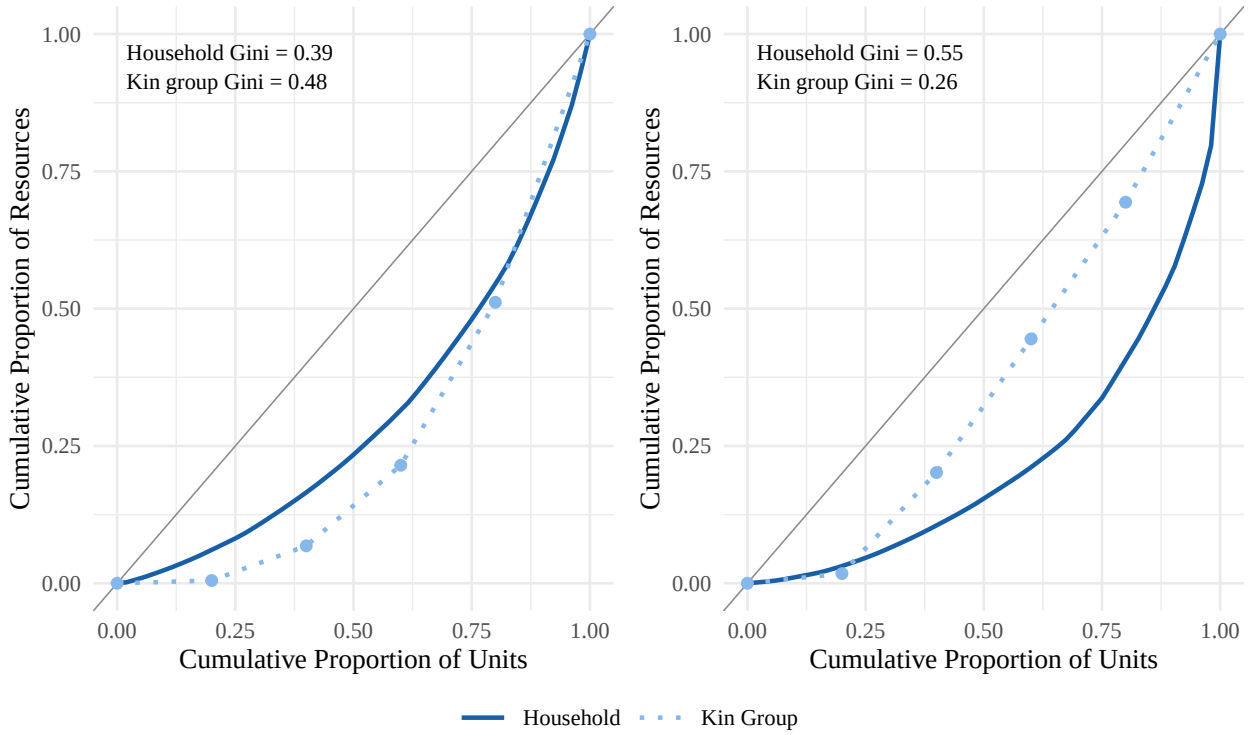
Appendix C1 holds nine households fixed, with units of resources (1, 1, 2, 2, 3, 4, 6, 8, and 12). Panel A shows these households on their own, with a Gini of 0.43. In Panels B to D, the same households form three kin groups of 2, 3, and 4 households (HH), changing only which households belong to which kin group. Each bar is a kin group, stacked with the households within it, and the kin group total is shown above it. In Panel B, high- and low-resource households are spread across kin groups of every size. Kin group totals are similar, and the kin group Gini falls to 0.07. In Panel C, high-resource households form the smallest kin group. The larger kin groups' extra members partly offset their lower resources, and the kin group Gini (0.24), which

remains below the household Gini. In Panel D, high-resource households form the largest kin group. Its total is high both because its members have more resources and because it has more members, so the kin group Gini (0.48) exceeds the household Gini. This example shows that, if high-resource households select into larger kin groups, then kin group inequality would be higher than household inequality.

Appendix C2: Plotting Lorenz Curves for Extreme Cases in Simulated Data

Panel A: Kin group Gini above household Gini

Panel B: Kin group Gini below household Gini

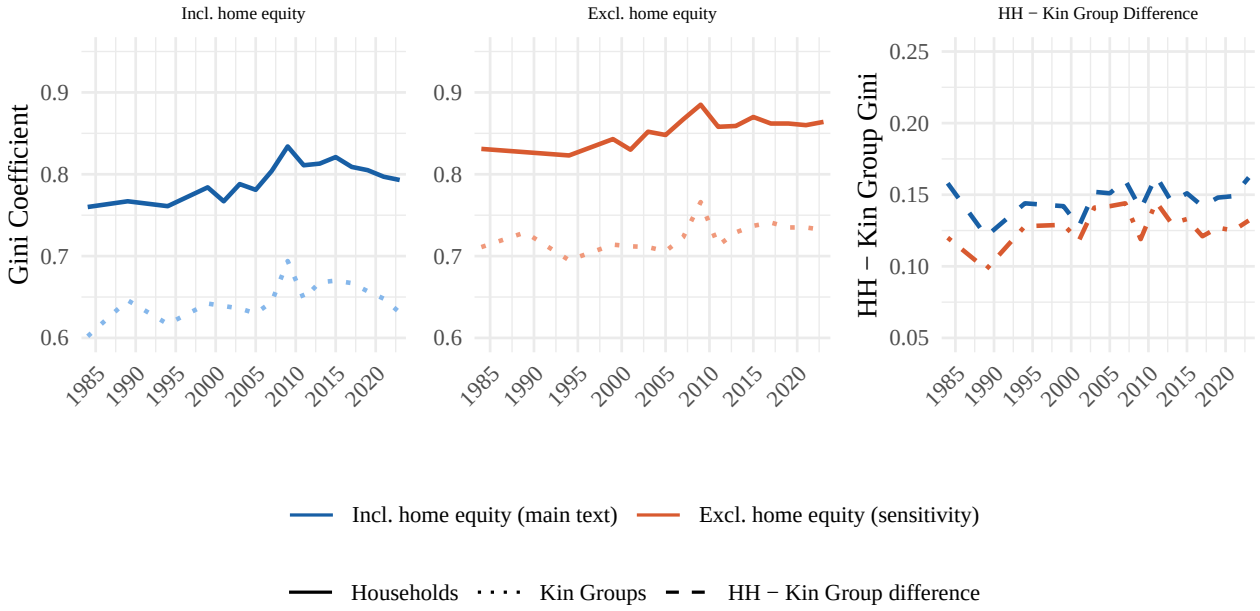


Appendix C2 extends this illustration with simulated data. In each of 100 simulations, we draw resources for 52 households independently from a lognormal distribution and assign them at random to five kin groups of 1, 6, 10, 15, and 20 households. We then compute the household Gini across the 52 households and the kin group Gini across the five kin group totals. The household Gini is higher in 95% of simulations, with a median difference of 0.11. Appendix C2 plots Lorenz curves for the two most extreme simulations. The further a curve is from the 45-degree line of perfect equality, the higher the Gini coefficient. Panel A resembles Panel D of Appendix C1. By chance, the larger kin groups drew higher-resource households, and the kin group Gini (0.48) exceeds the household Gini (0.39). Panel B resembles Panel B of Appendix C1. A small kin group drew several of the highest-resource households, which offsets its size, and kin group totals end up similar. Here the kin group Gini (0.26) is far below the household Gini (0.55).

Appendix D: Sensitivity Analyses — Gini Coefficients

Across a range of specifications, estimated inequality using the Gini is lower when kin groups are used as the unit of analysis instead of households. This pattern is not driven by race, nor does it disappear when expanding the sample, excluding home equity in measures of wealth, including negative wealth values, standardizing by household size, or applying weights. For each specification, results are recalculated and displayed alongside the main results.

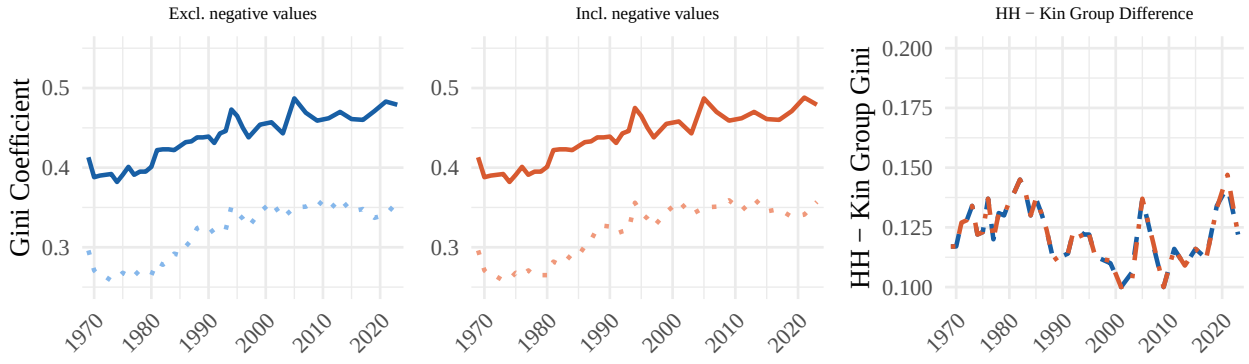
Appendix D1: Sensitivity — Wealth With and Without Home Equity



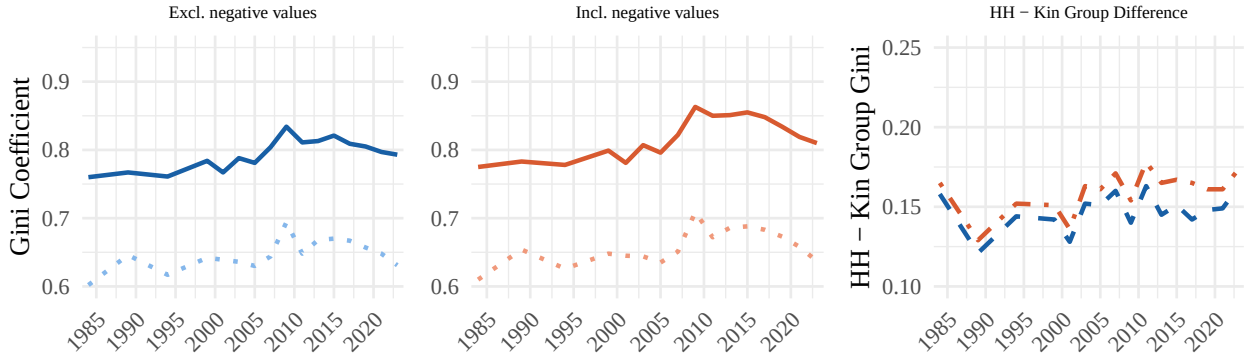
Results presented in the main text include home equity for measures of wealth. Appendix D1 shows Gini coefficients when home equity is excluded (middle graphs), and reports the difference between Gini coefficients at the household and kin group levels (graphs on the right). Excluding home equity makes estimated inequality lower and minimizes the observed difference between household and kin group Gini coefficients.

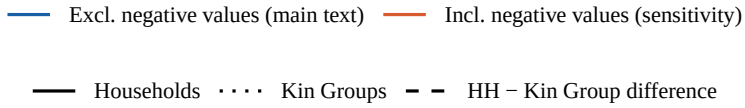
Appendix D2: Sensitivity — Negative Values

Panel A: Income



Panel B: Wealth

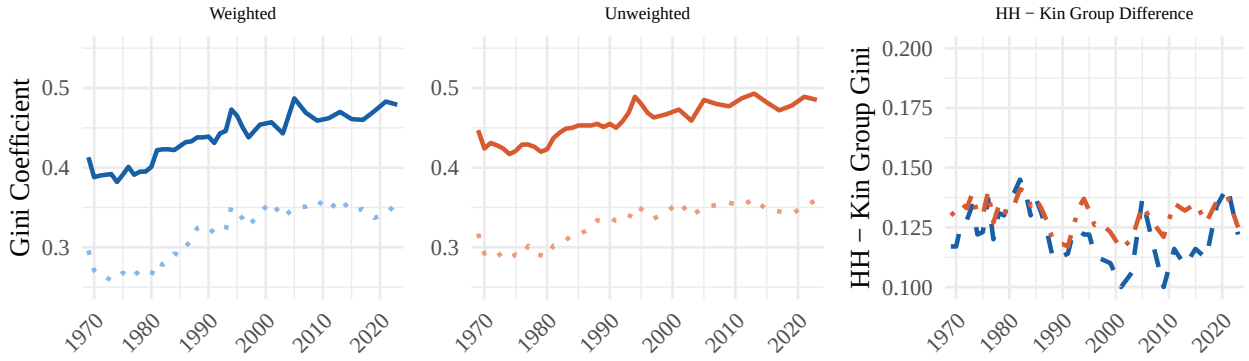




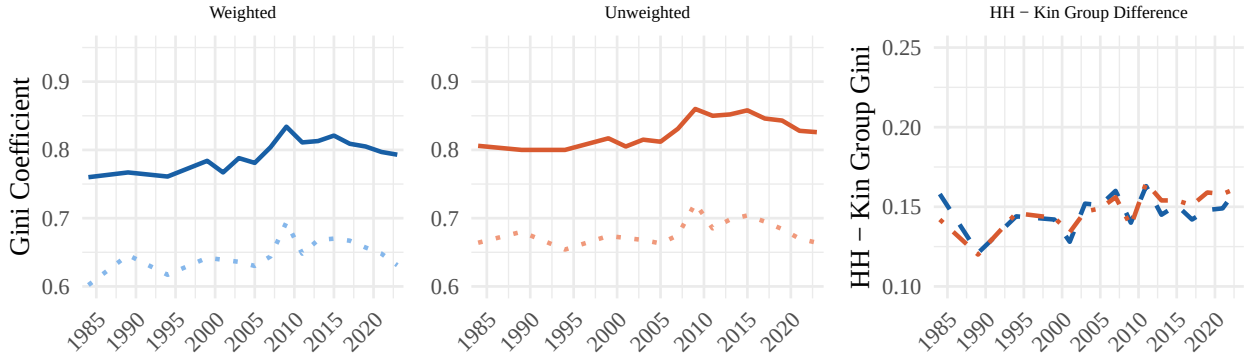
Results presented in the main text set negative values for household or kin group income and wealth to zero. Appendix D2 shows Gini coefficients when negative values are retained (middle graphs), and reports the difference between Gini coefficients at the household and kin group levels (graphs on the right). Negative income is a relatively rare event occurring when debt from family businesses or farms exceeds income. In our data, only 107 household-years and 6 kin group-years report negative income (0.04% of household-years and 0.01% of kin group-years). Having negative wealth is much more common: 14.69% of household-years and 5.73% of kin group-years report negative wealth. Overall, including negative values does not change estimated Gini coefficients at the household or kin group levels.

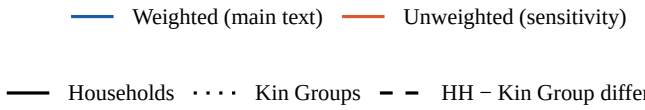
Appendix D3: Sensitivity — Weighting

Panel A: Income



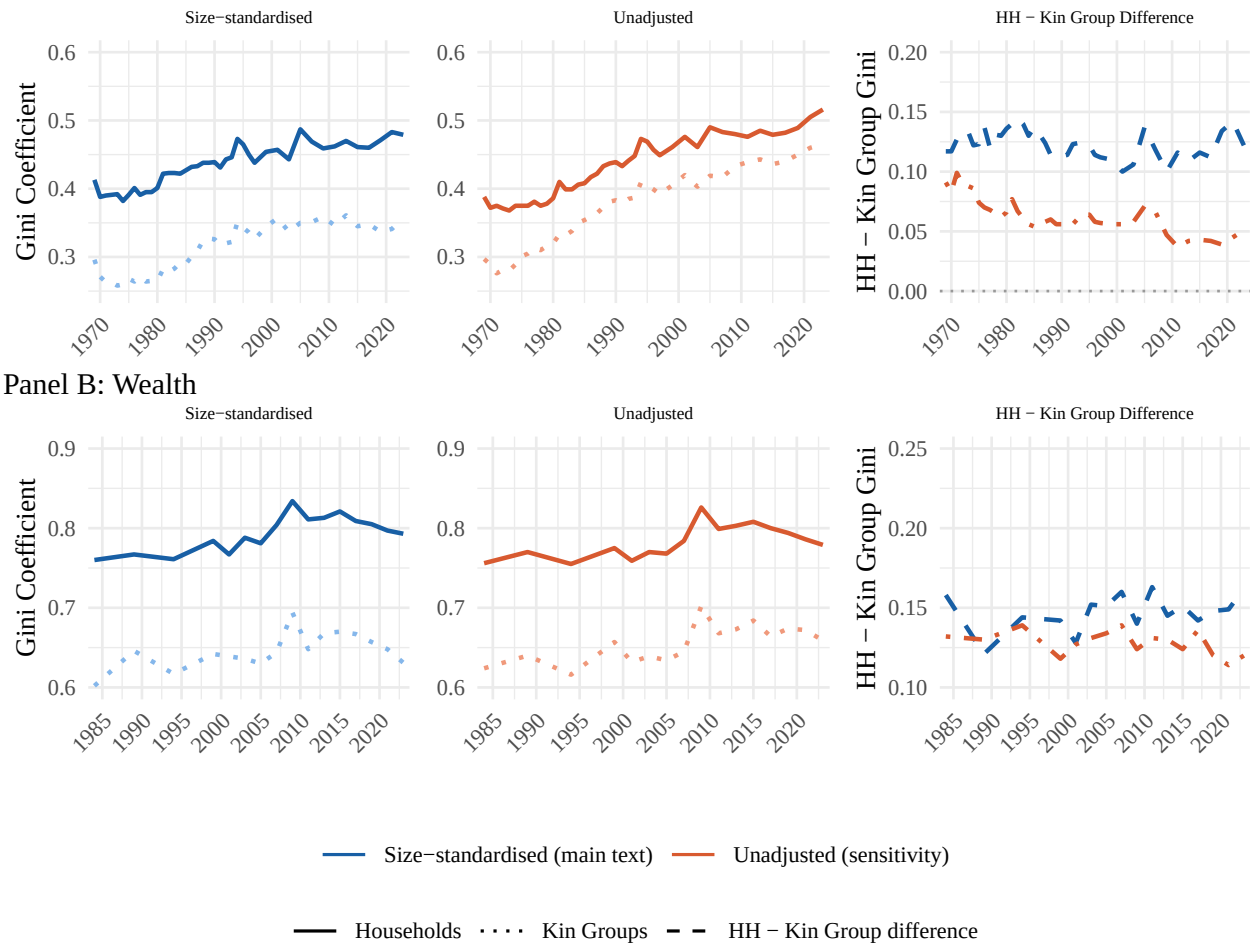
Panel B: Wealth





All results presented in the main paper are weighted. Household-level Gini coefficients are weighted using PSID-provided family weights, while kin group-level Gini coefficients are weighted using a constructed kin group weight based on the PSID family weight. Weighting does not substantially change the results.

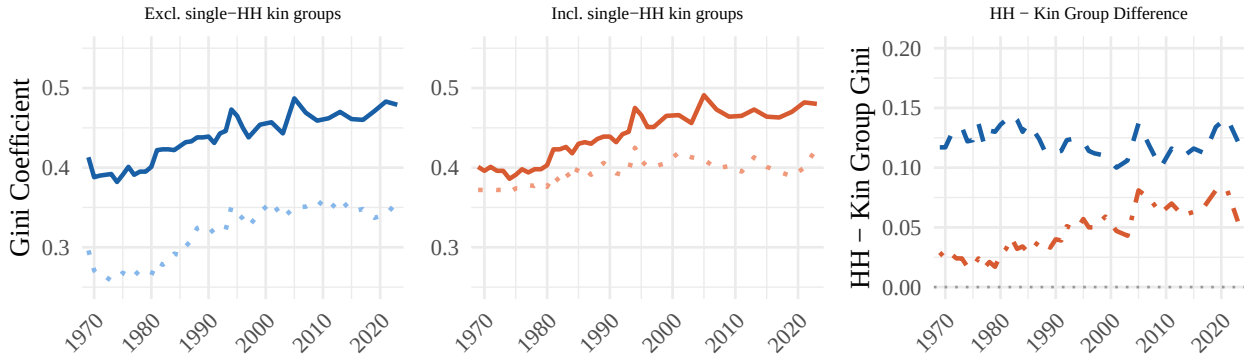
Appendix D4: Sensitivity — Size Standardization



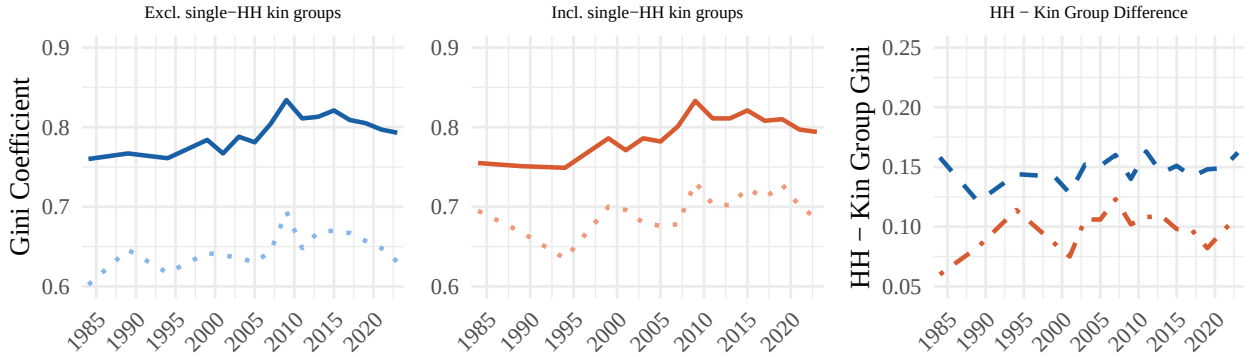
Results in the main text standardize income, wealth, and resulting Gini coefficients by household and kin group size. Estimated inequality is higher when results are not standardized by size and the gap between household and kin group Gini coefficients is smaller. However, Gini coefficients are higher at the household versus kin group level irrespective of whether measurements are adjusted by size.

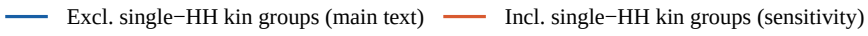
Appendix D5: Sensitivity — Single-Household Kin Groups

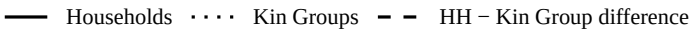
Panel A: Income



Panel B: Wealth





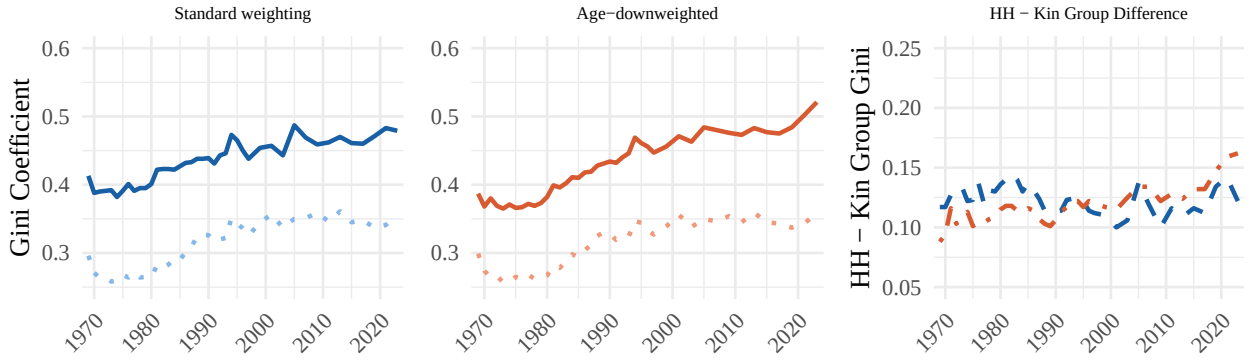


In reality, all households belong to kin groups that consist of multiple households. However, some households in the PSID do not have kin groups due to survey design and attrition. The survey is genealogical, meaning that initial households only have measurable kin groups once their descendants respond to the survey. Attrition also might limit responses from some households, thereby erasing kin groups for the households they are connected to.

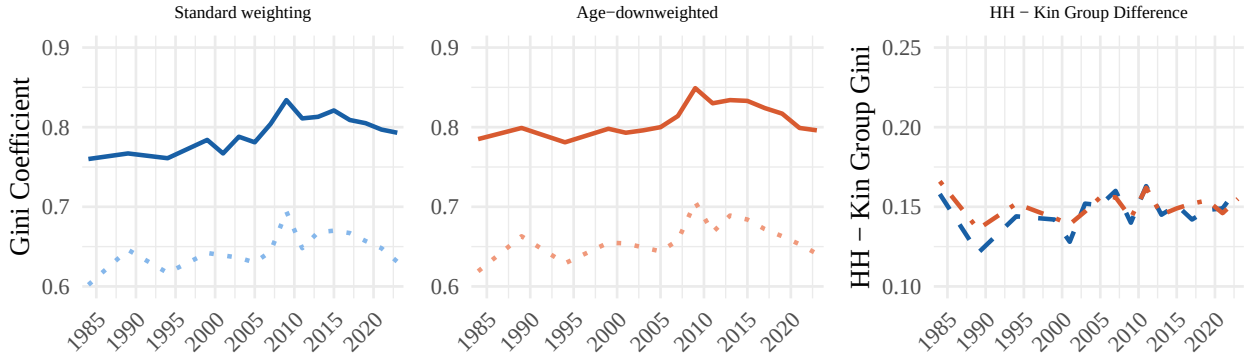
Because these limitations are imposed by the survey design, not social reality, we exclude household-years where households are alone in their kin group from both household and kin group measurements. The graphs on the left side show this. When including these single-household kin groups, or "kin groupless" households, the difference in Gini coefficients at the household and kin groups levels is minimized but not eliminated. The graphs in the middle show Gini coefficients when single-household kin groups are included. The graphs on the right side show the difference between household and kin group (HH - kin group) Gini coefficients for each group. Differences in estimated inequality at the kin group and household levels are smaller when including kin groupless households, however, they still exist and are increasing over time.

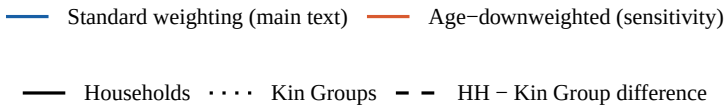
Appendix D6: Sensitivity — Age Weighting

Panel A: Income



Panel B: Wealth

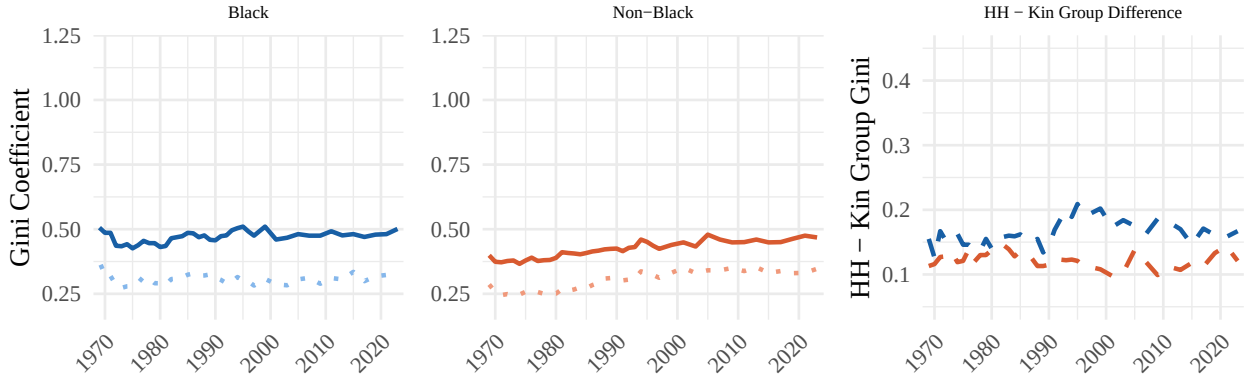




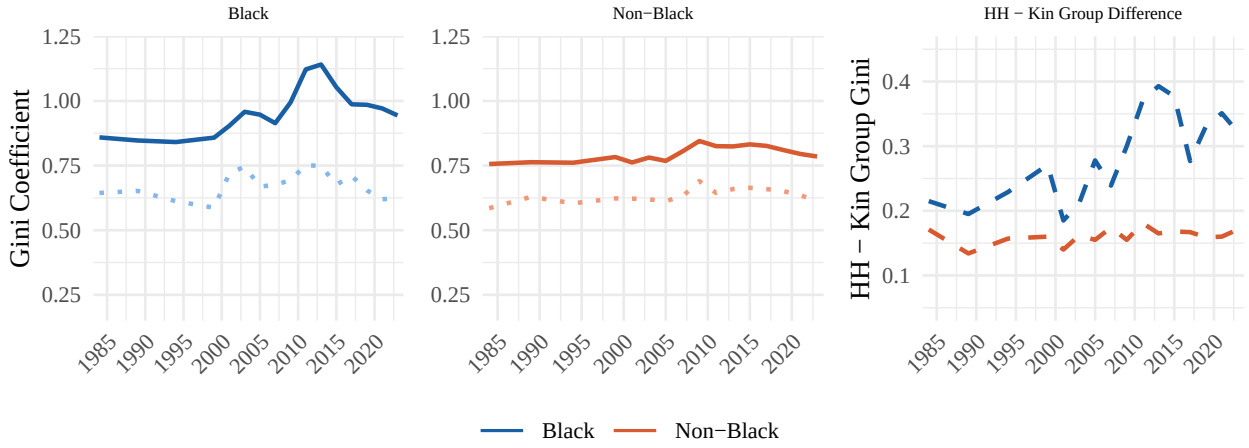
Results in the main text use standard PSID family and kin group weights. Appendix D6 shows Gini coefficients when these weights are downweighted for older households and kin groups (middle graphs): each weight is multiplied by the sample mean age divided by the unit's mean age. Gini coefficients remain higher at the household level than at the kin group level under age weighting.

Appendix D7: Gini Coefficients by Race Subgroup

Panel A: Income



Panel B: Wealth (incl. home equity)



Appendix D7 shows that the differences between Gini coefficients at the household and kin group level are not driven by racial subgroups. Graphs on the left side show Gini coefficients for Black households and kin groups. Graphs in the middle show Gini coefficients for non-Black households and kin groups. The graphs on the right side show the difference between household and kin group (HH - kin group) Gini coefficients for each group. Gini coefficients for income (Panel A) and wealth (Panel B) are higher at the kin group level for both Black and non-Black kin groups, indicating that inequality is lower when estimated at the kin group level across racial groups.

Appendix E: Sensitivity Analyses — Race Ratios

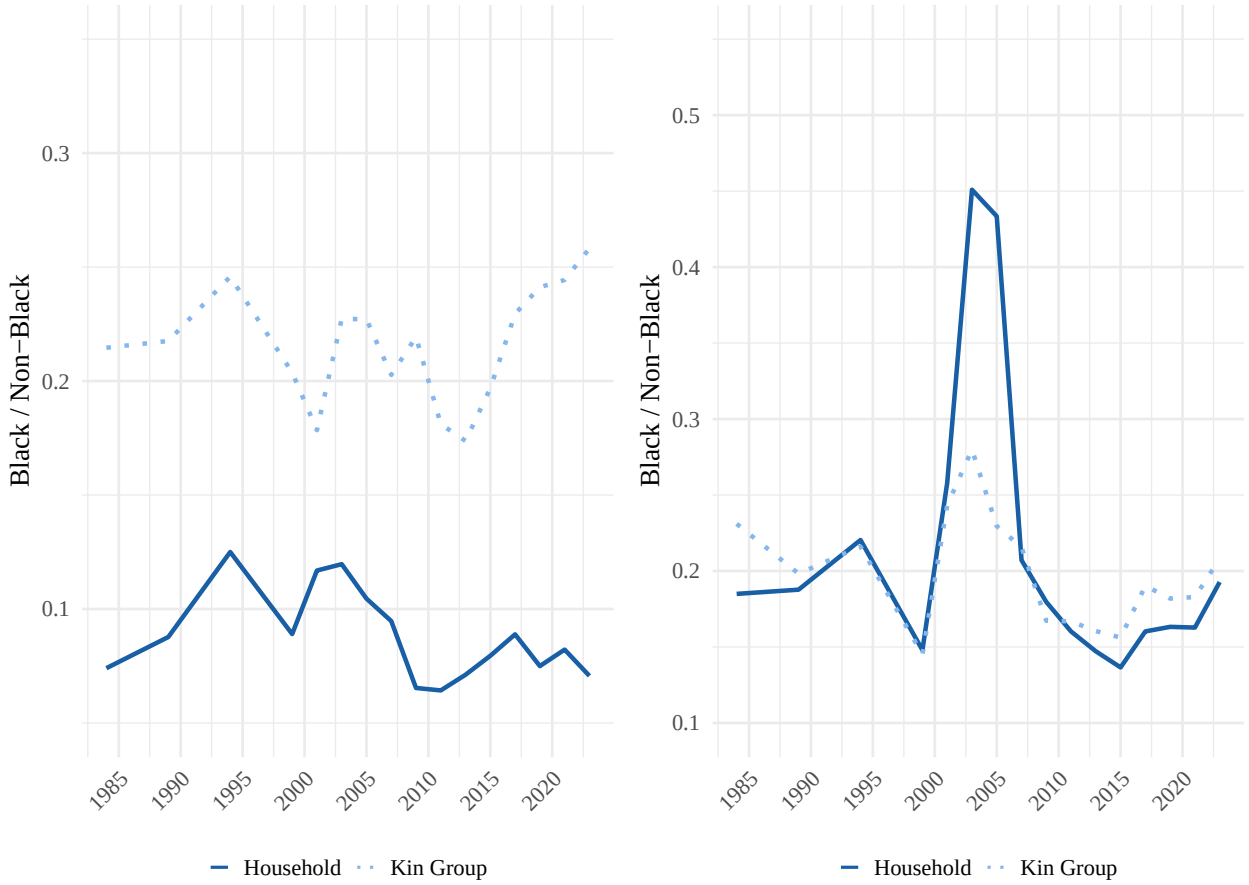
Racial wealth gaps estimated using mean, rather than median, wealth are about the same at the household and kin group levels. Appendix E1 plots the difference between the mean and median wealth ratios and explains why median racial wealth gaps are presented.

Across a range of specifications, estimated median racial wealth gaps are lower when kin groups are used as the unit of analysis instead of households. This pattern does not disappear when expanding the sample, excluding home equity in measures of wealth, including negative wealth values, standardizing by household size, or applying weights. For each specification, results are recalculated and displayed alongside the main results.

Appendix E1: Mean and Median Wealth Ratios

Panel A: Median wealth ratio

Panel B: Mean wealth ratio

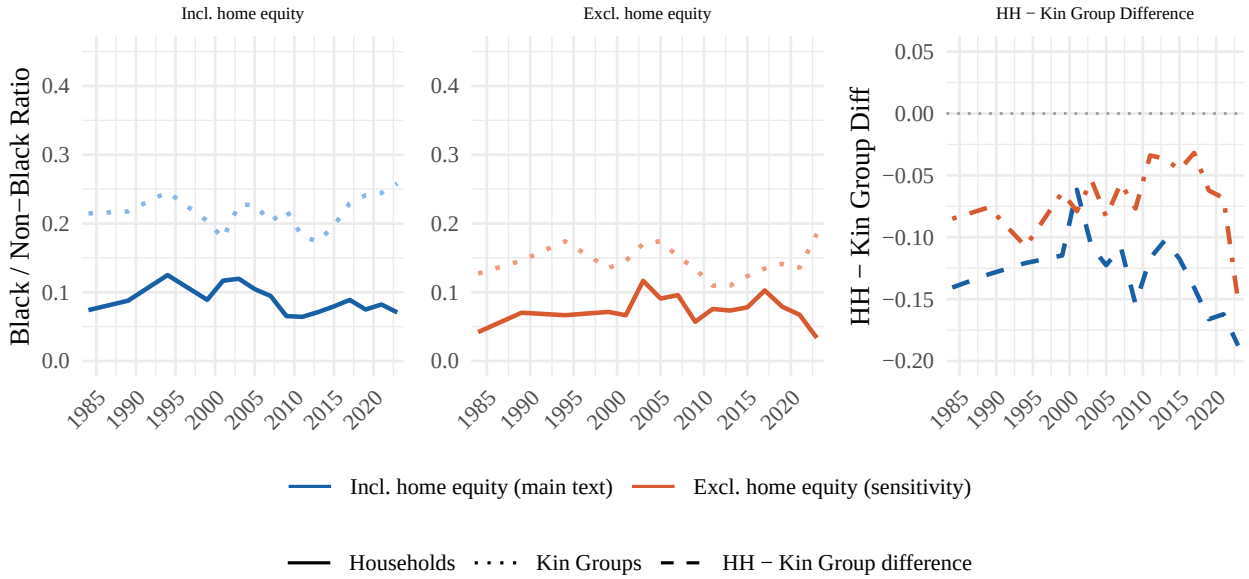


Appendix E1 reports Black / non-Black wealth ratios for both households and kin groups. Panel A reproduces the median wealth ratios (as in Figure 3 of the main paper), while Panel B presents mean wealth ratios. Using medians, the racial wealth gap differs between households and kin groups (Panel A), whereas using means, the two are about the same (Panel B).

Race ratios using mean wealth, when pooled across all years, show that Black households hold 19 cents for every dollar of wealth held by non-Black households, while Black kin groups hold 18 cents for every dollar held by non-Black kin groups. The convergence in mean-based estimates is driven by a small number of high-wealth Black households in the PSID, particularly in 2004–2005, which skew household-level averages. These results highlight the sensitivity of mean-based measures to outliers, which is why we present race ratios based on median wealth in the main text and in Appendices E2–E7.

Appendix E2: Race Ratios — Wealth With and Without Home Equity

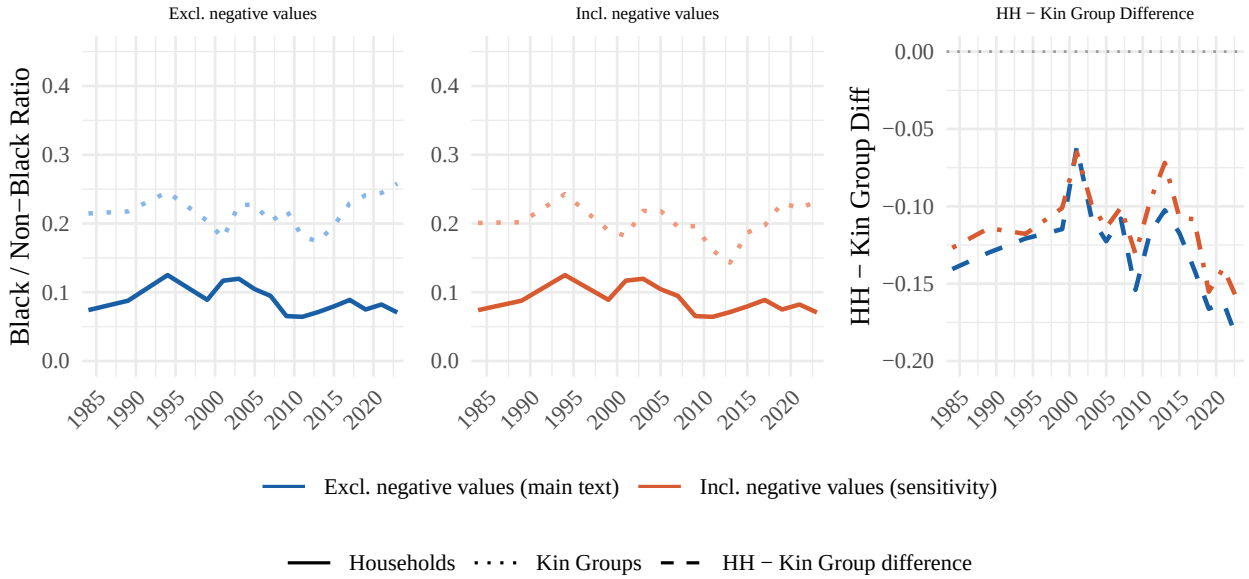
Median wealth ratio



Appendix E2 compares median Black / Non-Black wealth ratios when home equity is included (main specification, left panel) versus excluded (middle panel). The median racial wealth gap between kin groups, versus households, shows less inequality irrespective of whether wealth measures include home equity.

Appendix E3: Race Ratios — Negative Values

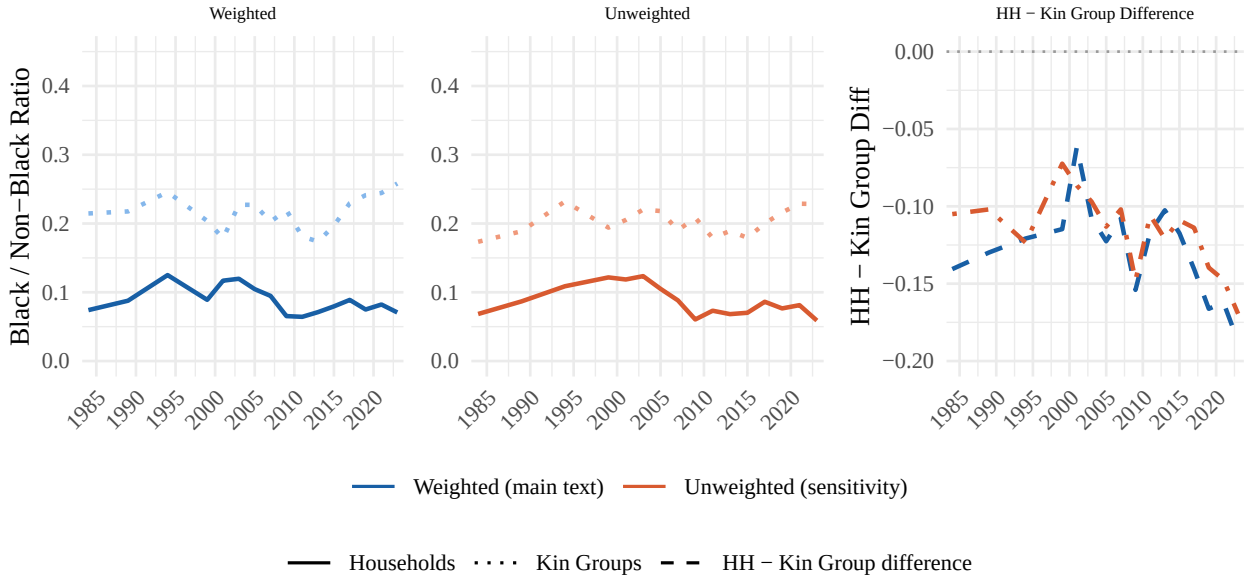
Median wealth ratio



Appendix E3 shows median Black / Non-Black wealth ratios when negative wealth values are excluded (main specification, left panel) versus included (middle panel). Including negative values does not change the estimated race ratios.

Appendix E4: Race Ratios — Weighting

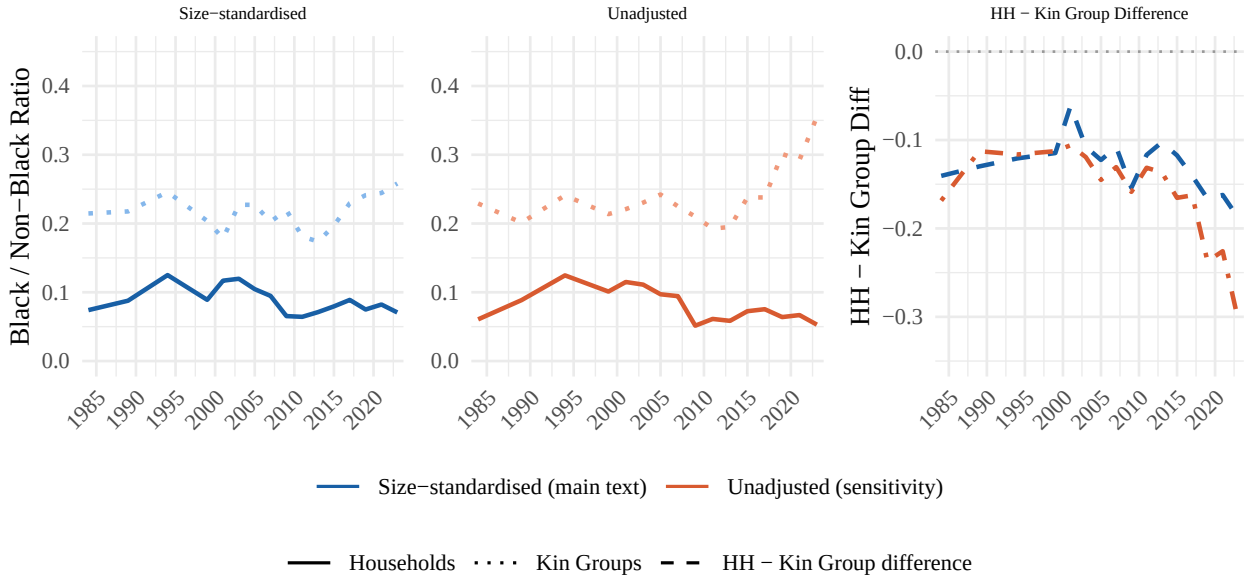
Median wealth ratio



Appendix E4 shows median Black / Non-Black wealth ratios using weighted estimates (main specification, left panel) versus unweighted estimates (middle panel). Weighting does not substantially change the estimated race ratios.

Appendix E5: Race Ratios — Size Standardization

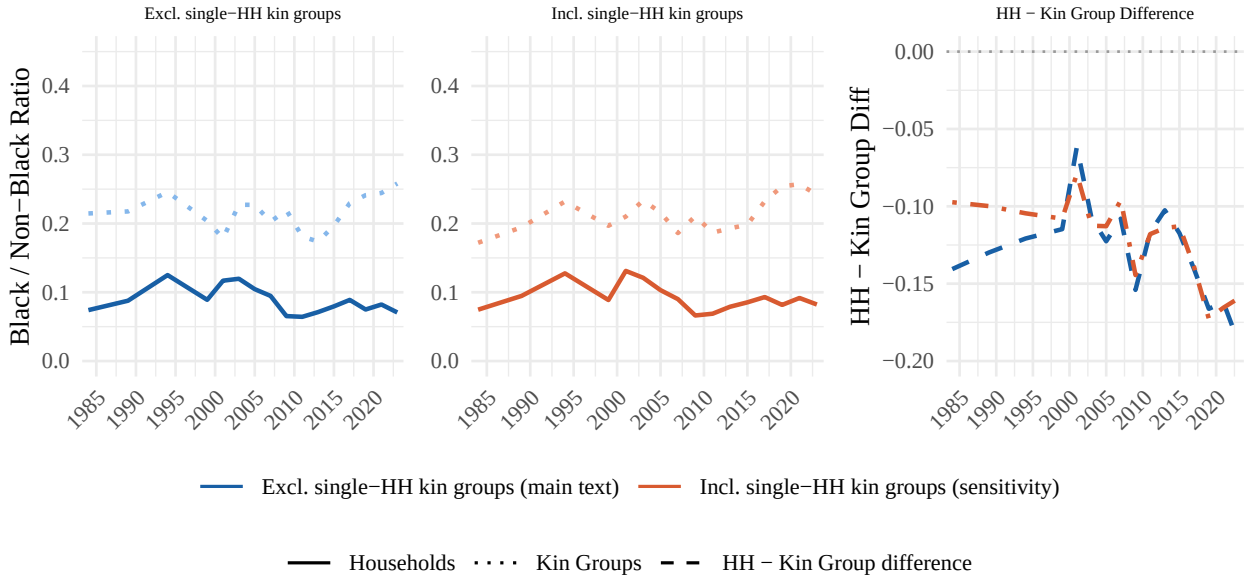
Median wealth ratio



Appendix E5 compares median Black / Non-Black wealth ratios when wealth is size-standardized by household and kin group size (main specification, left panel) versus unadjusted (middle panel). Standardizing by size does not substantially change the estimated median wealth race ratios from 1984–2015. From 2015–2023, the difference between median wealth gaps at the kin group versus household level widens.

Appendix E6: Race Ratios — Single-Household Kin Groups

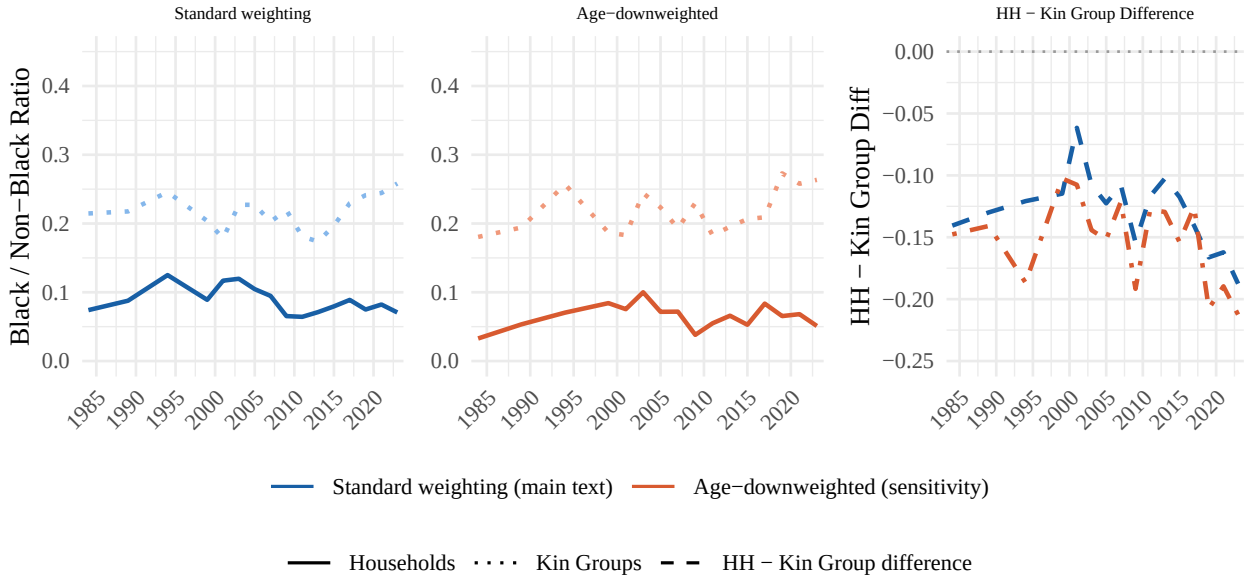
Median wealth ratio



Appendix E6 shows median Black / Non-Black wealth ratios when single-household kin groups are excluded (main specification, left panel) versus included (middle panel). Including single-household kin groups does little to change median racial wealth gaps (right panel). In particular, the median racial wealth gap between kin groups, versus households, shows less inequality irrespective of whether single-household kin groups are included.

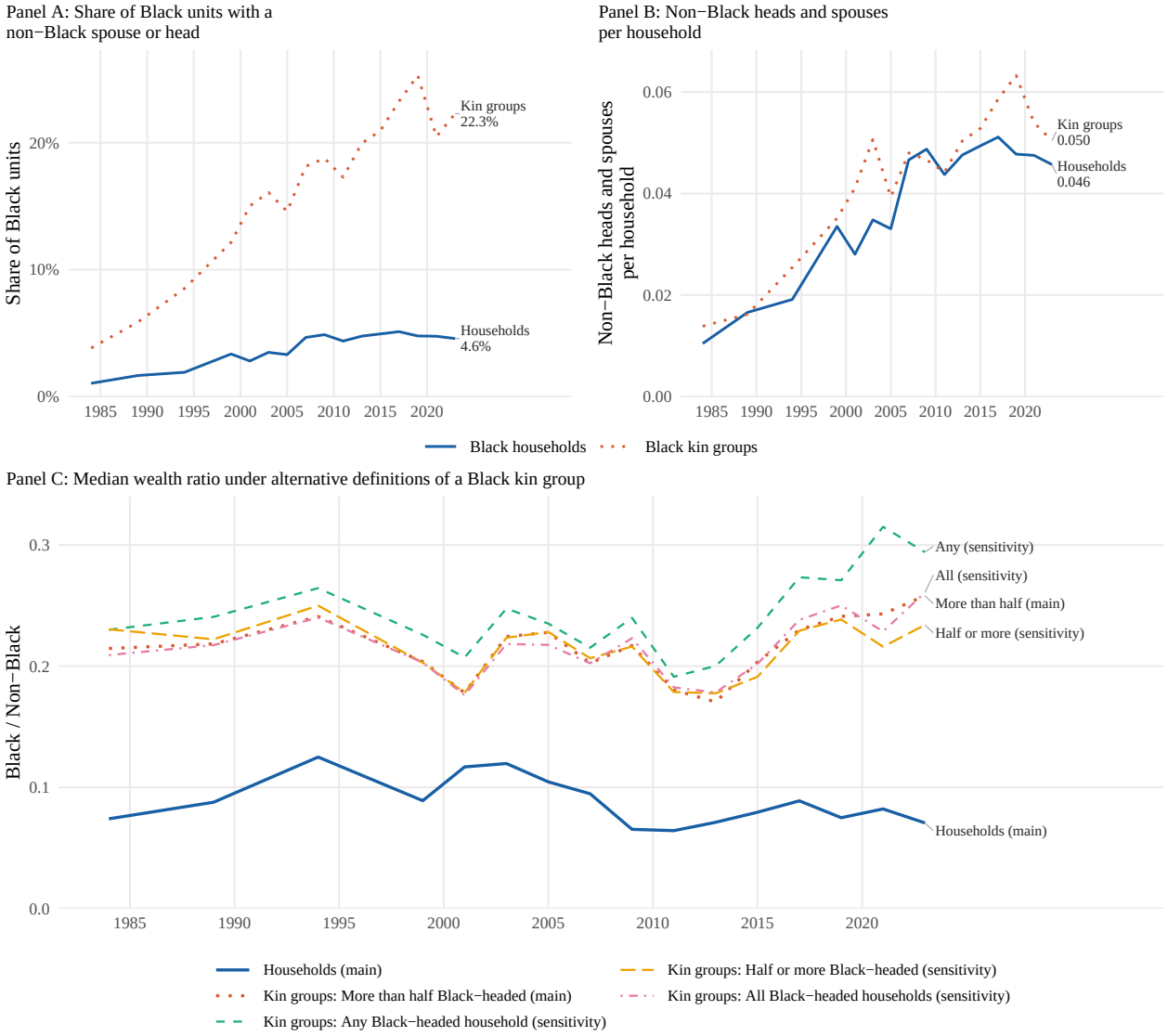
Appendix E7: Race Ratios — Age Weighting

Median wealth ratio



Appendix E7 shows median Black / Non-Black wealth ratios using standard weights (main specification, left panel) versus weights that downweight older households and kin groups (middle panel). Each weight is multiplied by the sample mean age divided by the unit's mean age.

Appendix E8: Racial Composition of Black Households and Kin Groups



Appendix E8 describes the racial composition of Black households and kin groups. Race is measured for heads and spouses only. Panel A shows the weighted share of Black households with a non-Black spouse and the weighted share of Black kin groups with at least one non-Black head or spouse. Panel B divides the number of non-Black heads and spouses in each kin group by the number of households within the kin group. All panels use the wealth sample (1984–2023). Panels A and B include 46,262 Black household-years and 8,608 Black kin group-years from 827 unique Black kin groups. Panel C reproduces Figure 3 under four definitions of a Black kin group. The number of Black kin group-years (7,789 to 9,397) and unique Black kin groups (814 to 972) does not vary substantially across definitions.

Because kin groups span several households, they are more likely than households to include non-Black individuals. In 2023, 22.3% of Black kin groups included at least one non-Black head or spouse, compared with 4.6% of Black households with a non-Black spouse (Panel A). Per household, however, the two are similar (0.050 vs. 0.046 non-Black heads and spouses per household; Panel B): the difference reflects the size of kin groups rather than higher rates of interracial partnership within them. Panel C shows that the median racial wealth gap is smaller for kin groups than for households whether a Black kin group is defined as having any, half or more, more than half (main specification), or all Black-headed households.

Appendix F: Household and Kin Group Size and Age

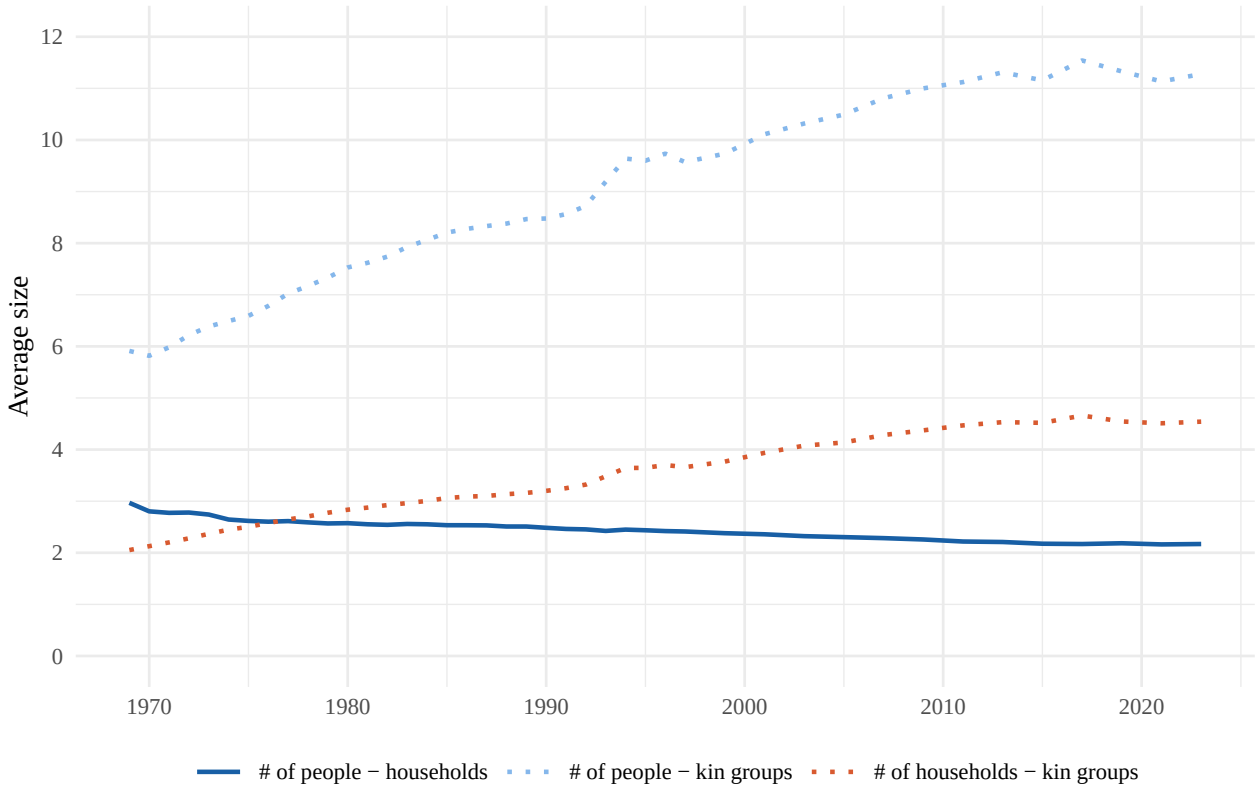
Appendix F1: Characteristics of Households and Kin Group Sample Members, 1984 and 2023

Appendix F1. Characteristics of Households and Kin Groups, 1984 and 2023

	1984		2023	
	Households	Kin Groups	Households	Kin Groups
Average age (years)	29.7	28.9	39.6	37.3
Average # of people	2.55	8.06	2.17	11.28
Average # of households	—	3.01	—	4.54
Median income	$67,610	$215,901	$68,280	$366,368
Median wealth	$67,451	$337,166	$131,000	$976,000
N (unit-years)	5,496	1,731	8,490	1,610
N Black (unit-years)	2,164	643	3,550	513

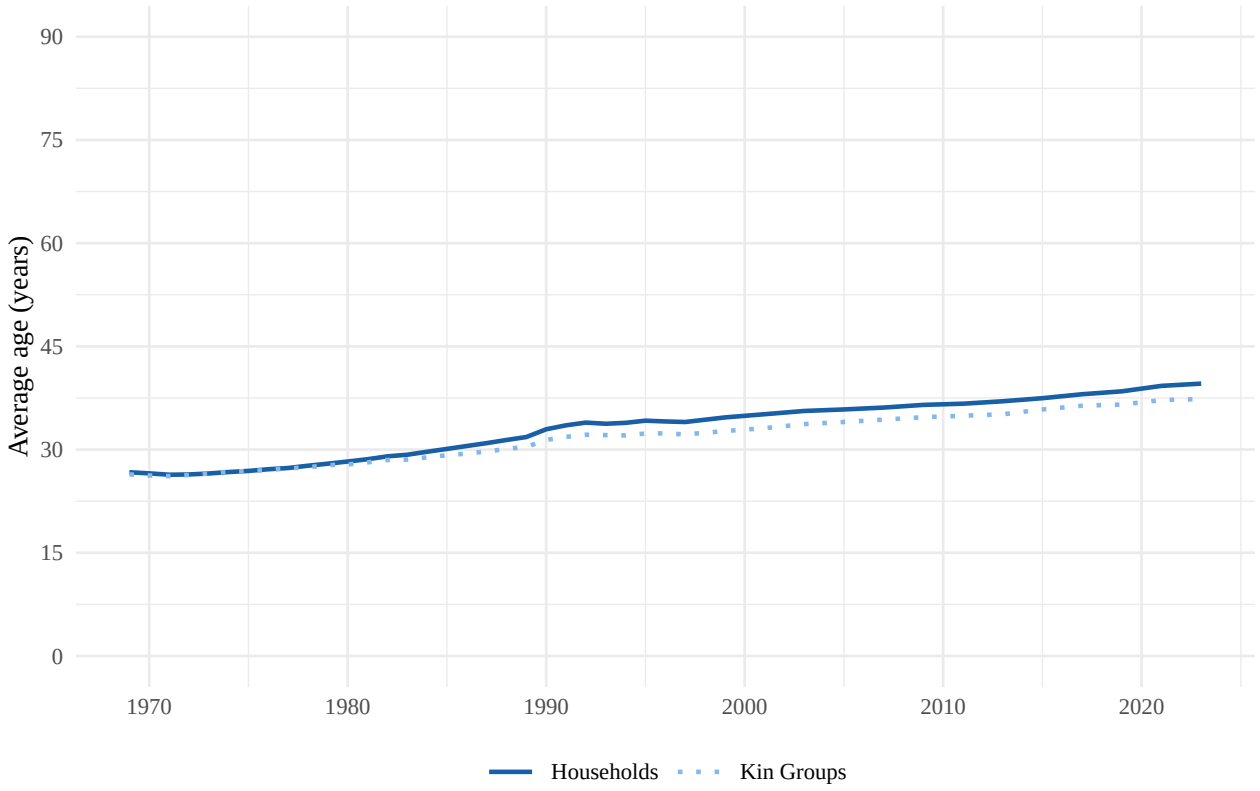
Note: Average age, number of people, and number of households are based on the income sample (1969-2023), which contains 2,906 unique kin groups. Median wealth is based on the wealth sample (1984-2023) and includes home equity. Size and median values are weighted using PSID family and constructed kin group weights (see Methods). Pooled across all years, the wealth sample contains 122,069 household-years and 26,655 kin group-years (46,287 and 8,470 Black, respectively) from 2,636 unique kin groups, with an average size of 2.27 individuals per household and 10.56 individuals (4.18 households) per kin group, and an average age of 36.3 for households and 34.1 for kin groups. Median income and wealth include households and kin groups with negative income or wealth. Estimates are weighted, adjusted for inflation, and expressed in 2023 dollars.

Appendix F2: Household and Kin Group Size Over Time



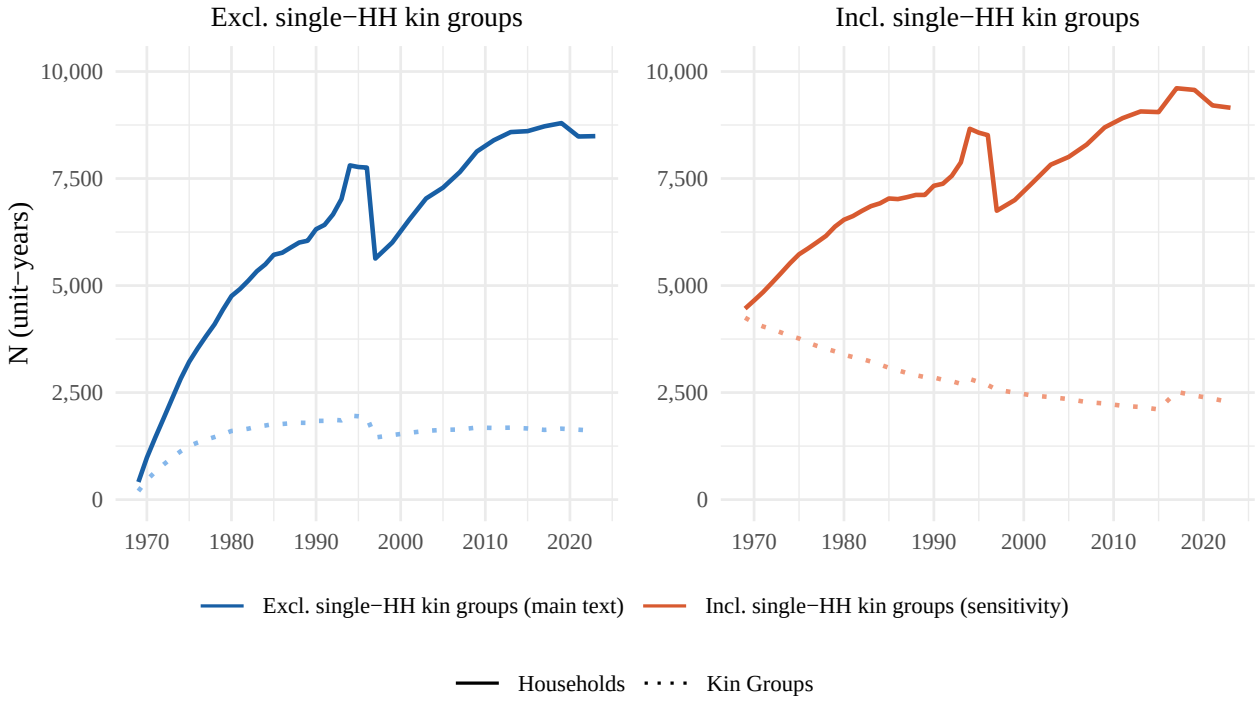
Appendix F2 plots the weighted average number of people per household and per kin group (blue), and the weighted average number of households per kin group (orange), in each year of the income sample.

Appendix F3: Average Age of Households and Kin Groups Over Time



Appendix F3 plots the average age of households and kin groups in each year of the income sample.

Appendix F4: Sample Size With and Without Single-Household Kin Groups



Appendix F4 plots the number of households and kin groups in each year when single-household kin groups are excluded (main specification, left panel) versus included (right panel).

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3. Wolff, E. The Size Distribution of Wealth in the United States: A Comparison among Recent Household Surveys. in *Wealth, Work, and Health: Innovations in Measurement in the Social Sciences* (Panel Study of Income Dynamics, 209 AD).
4. Pfeffer, F. T., Danziger, S. & Schoeni, R. F. Wealth Disparities before and after the Great Recession. *The Annals of the American Academy of Political and Social Science* **650**, 98–123 (2013).
5. Aaberge, R. Gini's nuclear family. *Journal of Economic Inequality* **5**, 305–322 (2007).